

CPEG495 Lab 3.5

By Furdeen Hasan

This Document Contains the Responses for Lab 3.5 Assignments: 1, 2, and 3.

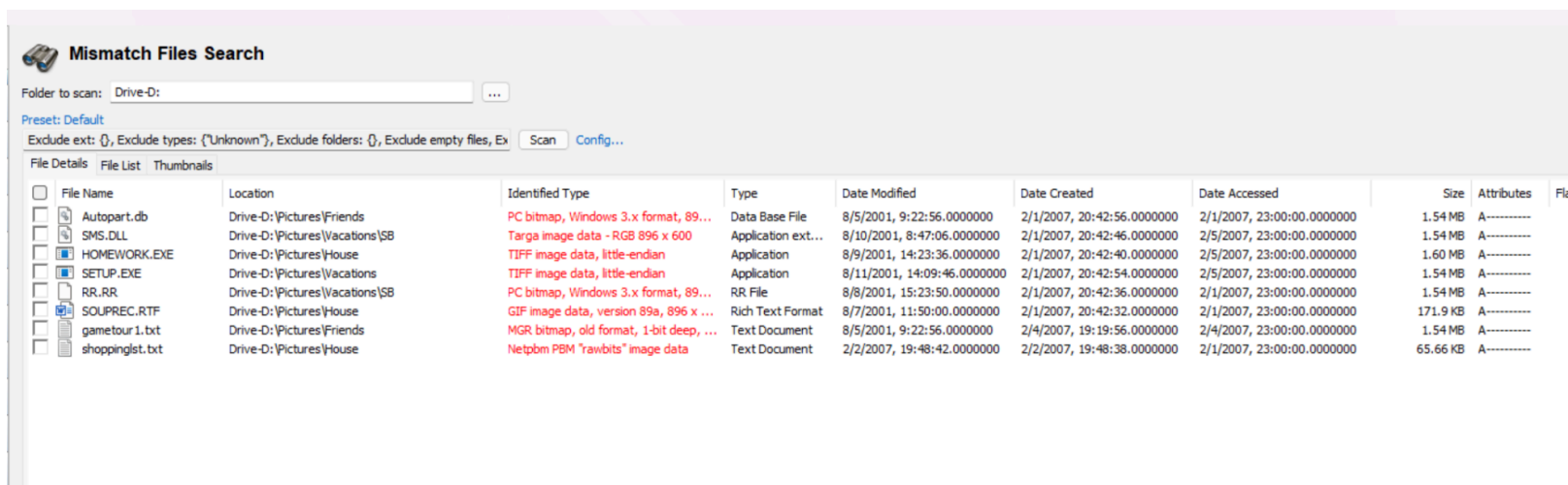
Lab Assignment 1:

1: Using OSForensics or Autopsy, search C08carve.dd for graphics files that were manipulated to appear like other file types.

Response:

Using OSForensics, I ran a Mismatch Files Search on the mounted forensic image (c08carve.dd). The tool identified 8 files with extensions that do not match their actual file types. These mismatches suggest attempts to conceal graphic evidence or malicious content under seemingly harmless file types. This is common in cybercrime investigations, especially for hiding contraband or secret communication within images.

Screenshot of Observations



File Name	Location	Identified Type	Type	Date Modified	Date Created	Date Accessed	Size	Attributes	File
☐ Autopart.db	Drive-D:\Pictures\Friends	PC bitmap, Windows 3.x format, 89...	Data Base File	8/5/2001, 9:22:56.0000000	2/1/2007, 20:42:56.0000000	2/1/2007, 23:00:00.0000000	1.54 MB	A-----	File
☐ SMS.DLL	Drive-D:\Pictures\Vacations\SB	Targa image data - RGB 896 x 600	Application ext...	8/10/2001, 8:47:06.0000000	2/1/2007, 20:42:46.0000000	2/5/2007, 23:00:00.0000000	1.54 MB	A-----	
☐ HOMEWORK.EXE	Drive-D:\Pictures\House	TIFF image data, little-endian	Application	8/9/2001, 14:23:36.0000000	2/1/2007, 20:42:40.0000000	2/5/2007, 23:00:00.0000000	1.60 MB	A-----	
☐ SETUP.EXE	Drive-D:\Pictures\Vacations	TIFF image data, little-endian	Application	8/11/2001, 14:09:46.0000000	2/1/2007, 20:42:54.0000000	2/5/2007, 23:00:00.0000000	1.54 MB	A-----	
☐ RR.RR	Drive-D:\Pictures\Vacations\SB	PC bitmap, Windows 3.x format, 89...	RR File	8/8/2001, 15:23:50.0000000	2/1/2007, 20:42:36.0000000	2/1/2007, 23:00:00.0000000	1.54 MB	A-----	
☒ SOUPREC.RTF	Drive-D:\Pictures\House	GIF image data, version 89a, 896 x ...	Rich Text Format	8/7/2001, 11:50:00.0000000	2/1/2007, 20:42:32.0000000	2/1/2007, 23:00:00.0000000	171.9 KB	A-----	
☐ gametour 1.txt	Drive-D:\Pictures\Friends	MGR bitmap, old format, 1-bit deep, ...	Text Document	8/5/2001, 9:22:56.0000000	2/4/2007, 19:19:56.0000000	2/4/2007, 23:00:00.0000000	1.54 MB	A-----	
☐ shoppinglist.txt	Drive-D:\Pictures\House	Netpbm PBM "rawbits" image data	Text Document	2/2/2007, 19:48:42.0000000	2/2/2007, 19:48:38.0000000	2/1/2007, 23:00:00.0000000	65.66 KB	A-----	

The Table Below Provides an analysis of the observed results:

File Name	Extension	Actual Type	Reason for Suspicion
Autopart.db	.db	PC bitmap image	Disguised image file; may contain hidden data
SMS.DLL	.dll	Targa image (RGB 896 × 600)	Should be a system file, but is actually an image
HOMEWORK.EXE	.exe	TIFF image	Fake executable, likely masking visual evidence
SETUP.EXE	.exe	TIFF image	Not an installer; masking image as executable
RR.RR	.rr	GIF image	Obscure extension, disguising a graphic
SOURCE.RTF	.rtf	Rich Text File	May still need analysis, depending on contents
gametour1.txt	.txt	MGR bitmap	Pretending to be plain text, actually an image
shoppinglist.txt	.txt	Netpbm PBM rawbits image	Hiding a graphic under a common filename

Lab Assignment 2:

Using the Mobilefish steganography tool [Links to an external site.](#), search for hidden messages in the graphics files that reveal some information about the overall crime (see 'Images' folder) Password is: "digiforensics".

Image:



Hidden Message: Party is off. Too many people know. Call party planners. 890-555-3367



Hidden Message: Kate and Sarah are coming at noon. Preparations are complete.



Hidden Message: Image in .jpg format, unable to decrypt



Hidden Message: The diamond is arriving at 8pm at the docks. Make sure you are ready for pick-up.



Hidden Message: Denise, the diamond is at the jewelry store. Steve will provide the value.



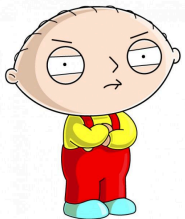
Hidden Message: Ms. Robinson, the watch is complete. Steven will love it. Stop by the jewelry store tomorrow.



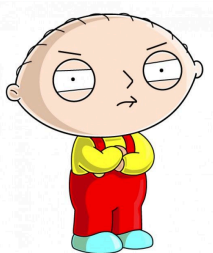
Hidden Message: Does your husband have any idea of our plans? I am sure that John doesn't know a thing. I can't wait.



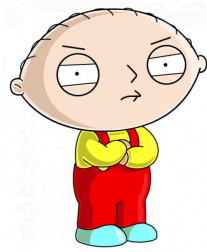
Hidden Message: Buyer contact- 475-555-0120 Proposed meeting at Valentino's on Thursday.



Hidden Message: Steven is coming home early. The diamond still needs to be analyzed. Call Moe to stall.



Hidden Message: Jack and Joe found Rico at the train station. Call the cleaner 647-555-9843



Hidden Message: Sensitive data related to crime



Hidden Message: Ariba Ariba - run away from this one...not relevant.



Hidden Message: The account number is 4567-8346-2918-0392. Payment for package.



Hidden Message: Rico has gone missing. This means trouble for us. Have to call the cleaner.



Hidden Message: Denise, the package hasn't arrived. Check with Rico.

Lab Assignment 3:

3: Using a mail client of your choice, analyze the emails provided to identify information related to the crime (HINT: Emails in MBOX format).

MBOX files inspected using: <https://www.mbox-viewer.com/>

Crime Message Invites:

First Batch (Duane Wilson to Jeff Avery) :



1. Encoded message hinting at a crime: Duane Wilson uses an encoded format and refers to the crime while addressing Jeff Avery.



2. Jeff Avery agrees with keeping things encoded to avoid being caught.

Subject: Re: One more message
To: Jeff Avery (jeffkavery@gmail.com)
From: Duane Wilson <radcon16@gmail.com>
Date: 2017-03-21T02:48:45.000Z

Thanks Doc of deception!

On Mar 20, 2017 10:47 PM, "Jeff Avery" <jeffkavery@gmail.com> wrote:

> Yessir, have to keep it secret to avoid getting caught in the open.

> On Mon, Mar 20, 2017 at 22:45 Duane Wilson <radcon16@gmail.com> wrote:

3. Duane appreciates Jeff's compliance: Duane replies positively to Jeff's willingness and discretion.

Second Batch: Jeff's Initial Crime Invitation and Response

Subject: Crime message 1
To: Jeff Avery (jeffkavery@gmail.com)
From: Duane Wilson <mr.duanewilson@gmail.com>
Date: 2017-03-21T02:05:20.000Z

Hey Jeff -

Interested in committing this crime? Say yes :-)

Prof

4. Initial invitation to commit a crime sent by Duane Wilson to Jeff Avery, indicating active recruitment.

Subject: Re: Crime message 1

To: Duane Wilson (mr.duanewilson@gmail.com)

From: Jeff Avery <jeffkavery@gmail.com>

Date: 2017-03-21T02:06:10.000Z

```
*Yesterday
*Everyday
*Sometimes

On Mon, Mar 20, 2017 at 10:05 PM Duane Wilson <mr.duanewilson@gmail.com>
wrote:

> Hey Jeff -
>
> Interested in committing this crime? Say yes :-))
>
> Prof
>
```

5. Jeff responds with hidden acronym: Jeff Avery responds with ambiguous wordplay, possibly coded affirmation using emphasized letters (Y, E, S).

Subject: Re: Crime message 1

To: Duane Wilson (mr.duanewilson@gmail.com)

From: Jeff Avery <jeffkavery@gmail.com>

Date: 2017-03-21T02:06:48.000Z

```
AM I EVER!
```

On Mon, Mar 20, 2017 at 10:06 PM Jeff Avery <jeffkavery@gmail.com> wrote:

```
> *Y*esterday
> *E*veryday
> *S*ometimes
>
> On Mon, Mar 20, 2017 at 10:05 PM Duane Wilson <mr.duanewilson@gmail.com>
```

6. Jeff confirms participation: Jeff Avery confirms willingness to participate in the crime with the response “AM I EVER!”.

Third Batch: Recruiting Natasha Wilson

Subject: Crime message 2

To: Natasha Wilson (nnlambert@gmail.com)

From: Duane Wilson <mr.duanewilson@gmail.com>

Date: 2017-03-21T02:06:08.000Z

Hey Natasha - recruiting for this crime I'm committing. Are you interested?

say YES please!!

Prof

7. Duane Wilson attempts to recruit Natasha Wilson into the crime in a separate message.

Subject: Re: Crime message 2

To: Duane Wilson (mr.duanewilson@gmail.com)

From: Natasha Wilson <nnlambert@gmail.com>

Date: 2017-03-21T02:22:08.000Z

Dear Dr. Wilson,

It has come to our attention that your password is about to expire in 5 days. Please go to the following link to update your password
<https://www.secureit.com>.

You will be asked to provide your current password and then you will also provide a new password. Your new password should include at least:

- 6-10 characters long
- one alpha character
- one numeric character
- one capital letter
- one special character

8. Natasha sends a fake phishing reply: Natasha replies with what appears to be a suspicious phishing-style message posing as a password reset.

Subject: Re: Crime message 2

To: Natasha Wilson (nnlambert@gmail.com)

From: Duane Wilson <mr.duanewilson@gmail.com>

Date: 2017-03-21T02:23:27.000Z

This looks like spam! Should I click on the link?

On Mon, Mar 20, 2017 at 10:22 PM, Natasha Wilson <nnlambert@gmail.com> wrote:

> Dear Dr. Wilson,
>
> It has come to our attention that your password is about to expire in 5
> days. Please go to the following link to update your password
> <https://www.secureit.com>.
>

9. Duane Wilson expresses suspicion about the legitimacy of the password reset message — possibly an attempted social engineering test or spam awareness tactic.

Fourth Batch: Recruiting Alex Luy

Subject: Mi Hermano

To: Alex Luy (alex.luy@gmail.com)

From: Duane Wilson <mr.duanewilson@gmail.com>

Date: 2017-03-21T02:07:19.000Z

Trying to recruit to get this cash homey. You in?

Prof

10. Duane Wilson contacts Alex Luy, attempting to recruit him into a money-related operation using casual slang to mask illegal intent.

Subject: Re: Mi Hermano

To: Duane Wilson (mr.duanewilson@gmail.com)

From: Alex Luy <alex.luy@gmail.com>

Date: 2017-03-21T02:09:01.000Z

Sure. How so?

On Mar 20, 2017 10:07 PM, "Duane Wilson" <mr.duanewilson@gmail.com> wrote:

> Trying to recruit to get this cash homey. You in?
>
> Prof
>

11. Alex Luy shows interest and asks for details, indicating willingness to participate.

Subject: Re: Mi Hermano

To: Alex Luy (alex.luy@gmail.com)

From: Duane Wilson <mr.duanewilson@gmail.com>

Date: 2017-03-21T02:09:43.000Z

I got you bro - details will come via the stego channel! Per usual.

On Mon, Mar 20, 2017 at 10:09 PM, Alex Luy <alex.luy@gmail.com> wrote:

> Sure. How so?
>

12. Duane replies that further details will be delivered through “the stego channel,” a reference to steganography — suggesting use of hidden messages in images.

Subject: Re: Mi Hermano

To: Duane Wilson (mr.duanewilson@gmail.com)

From: Alex Luy <alex.luy@gmail.com>

Date: 2017-03-21T02:19:49.000Z

dame dinero amigo

On Mar 20, 2017 10:09 PM, "Duane Wilson" <mr.duanewilson@gmail.com> wrote:

> I got you bro - details will come via the stego channel! Per usual.

>

> On Mon, Mar 20, 2017 at 10:09 PM, Alex Luy <alex.luy@gmail.com> wrote:

13. Alex confirms interest in the operation with the phrase “dame dinero amigo” (give me money, friend), implying motive and agreement to participate.

Fifth Batch: Alternate Email and Spanish Coordination

Subject: Hermano

To: Alex Luy (alex.luy@gmail.com)

From: Duane Wilson <radcon16@gmail.com>

Date: 2017-03-21T02:17:49.000Z

Psst...trying another account to plan the crime. What do you say bro?

Maybe respond in Espanol so they don't know what we are planning!

- dw

14. Duane tries to hide plans by switching accounts and suggests using Spanish.

Subject: Re: Hermano
To: Duane Wilson (radcon16@gmail.com)
From: Alex Luy <alex.luy@gmail.com>
Date: 2017-03-21T02:20:24.000Z

okay. que vamos a hacer. Vamos a robar informacion.

On Mar 20, 2017 10:17 PM, "Duane Wilson" <radcon16@gmail.com> wrote:

> Psst...trying another account to plan the crime. What do you say bro?

> Maybe respond in Espanol so they don't know what we are planning!

>

> - dw

>

15. Alex replies in Spanish to hide plans: Alex confirms the plan and suggests stealing information.

Email Analysis Summary:

The recovered MBOX email data reveals a coordinated effort involving Duane Wilson, Jeff Avery, Natasha Wilson, and Alex Luy in the planning and recruitment for a cybercrime. Duane initiates contact with each individual, using varying degrees of code, slang, and even multiple email addresses to conceal the intent of the operation. Jeff Avery not only agrees to participate but also suggests using encoded messages and expresses excitement. Natasha Wilson responds with a fake phishing-style message, possibly to mask her intent or as a decoy. Alex Luy, approached through a secondary email and encouraged to respond in Spanish, shows enthusiastic agreement and engagement in the planning, even requesting money. The repeated use of code words, Spanish, steganography, and informal language indicates an attempt to obscure the crime's details and evade detection.